

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester II Examination, 2023-24

Paper No. MTB-203A: Ancillary Mathematics I

Time: 3 Hours

Maximum Marks: 70

Ref. No.: 074894

*Instructions: Answer **five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks.*

Question 1. (Compulsory — 2 marks each.)

- (a) Draw the graph of $f(x) = |x|$ for $x \in \mathbb{R}$.
- (b) If $(x + 1, y - 2) = (3, 1)$, find x and y .
- (c) Let $A = \{1, 2, 3\}$, $B = \{3, 4\}$, $C = \{4, 5, 6\}$. Find $A \cup (B \cap C)$.
- (d) Let $A = \{1, 2\}$ and $B = \{3, 4\}$. Find the number of relations from A to B .
- (e) Let $V = \{a, e, i, o, u\}$ and $B = \{a, i, k, u\}$. Find $V - B$ and $B - V$.
- (f) Show that $A \cup B = A \cap B$ implies $A = B$.
- (g) Define diagonal matrix and give an example.
- (h) For the 3×3 matrix A with rows $(1, 2, 3)$, $(1, 3, 2)$, $(2, 4, 7)$, find the minor M_{12} .
- (i) Find a, b, c, d from the equation $\begin{pmatrix} 2a+b & a-2b \\ 5c-d & 4d+3b \end{pmatrix} = \begin{pmatrix} 4 & -3 \\ 11 & 24 \end{pmatrix}$.
- (j) Given matrices A with rows $(3, 2, 3)$, $(1, -5, 3)$, $(2, 2, -2)$ and B with rows $(6, 7, -5)$, $(-2, 2, -6)$, $(0, 1, 7)$, find $A + B$.
- (k) Find AB where $A = \begin{pmatrix} 1 \\ -3 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & -6 & 3 \end{pmatrix}$.

Question 2.

- (a) Show that the relation $R = \{(m, n) : |n - m| \text{ is even}\}$ on $X = \{3, 4, 5, 6\}$ is an equivalence relation. [6]
- (b) Find the adjoint of $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & -5 & 1 \\ 1 & 2 & 3 \end{pmatrix}$. [6]

Question 3.

(a) If $A = \begin{pmatrix} 1 & 3 & 3 \\ 3 & -2 & 1 \\ 4 & 1 & -1 \end{pmatrix}$, show $A^3 - 23A - 40I = O$. [6]

(b) Determine the row-rank of $A = \begin{pmatrix} 1 & 2 \\ 2 & 3 \\ 1 & 1 \end{pmatrix}$. [6]

Question 4.

(a) Find the rank of $A = \begin{pmatrix} 1 & 2 & -1 & 4 \\ 2 & 4 & 3 & 6 \\ -1 & -2 & 6 & -7 \end{pmatrix}$. [6]

(b) Solve the system $x + y + z = 6$, $y + 3z = 11$, $x - 2y + z = 0$. [6]

Question 5.

(a) Find the inverse of $A = \begin{pmatrix} 2 & -1 & 4 \\ 1 & 2 & 4 \\ 5 & 1 & 2 \end{pmatrix}$. [6]

(b) Examine the consistency of $x + 2y = 2$, $2x + 4y = 3$. [6]

Question 6.

(a) Find minors and cofactors of all elements of the determinant $\Delta = \begin{vmatrix} 2 & -3 & 5 \\ 6 & 0 & 4 \\ 1 & 5 & -7 \end{vmatrix}$. [6]

(b) Show that $R = \{(5, 5), (5, 3), (2, 2), (2, 4), (3, 5), (3, 3), (4, 2), (4, 4)\}$ on $A = \{2, 3, 4, 5\}$ is an equivalence relation. [6]

Question 7.

(a) Find the inverse of $A = \begin{pmatrix} 2 & -1 & 3 \\ -1 & 3 & -2 \\ 2 & -4 & 0 \end{pmatrix}$ by row operations. [6]

(b) Evaluate $\Delta = \begin{vmatrix} 1 & 2 & 4 \\ -1 & 3 & 0 \\ 4 & 1 & 0 \end{vmatrix}$. [6]

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